

Task 4: Hypothesis Test

$$\xi_{11}=904$$

$$\xi_{12}=62.3$$

$$\xi_{13}=2$$

$$\xi_{14}=922, 915, 911, 894, 923, 919, 927, 911, 896, 894$$

$\mu_0 = \xi_{11} = 904$ g, $\sigma_0 = \xi_{12} = 62.3$ g. Sample $N=10$ observations in ξ_{14} .

Question: Does the new system make higher weights? ($\xi_{13}=2$)

The task is based on the provided parameters:

- Old system average weight ($\mu_0 = \xi_{11}$): 904 g.
- Old system standard deviation ($\sigma_0 = \xi_{12}$): 62.3 g.
- Test Condition (ξ_{13}): 2 (Does the new system make higher weights?).
- New sample weights (ξ_{14}): 922, 915, 911, 894, 923, 919, 927, 911, 896, 894 ($N=10$).

```
#import libraries
import numpy as np
from scipy import stats as st
import matplotlib.pyplot as plt
```

Step 1:

We are testing whether the mean weight produced by the new system (μ_{new}) is significantly higher than the established mean $\mu_0 = 904$ g.

Model Proposal and Assumptions:

Model and Parameters (Old System)

We model the weight of hammers W based on the long period of observation as a Normal (Gaussian) distribution $W \sim N(\mu_0, \sigma_0^2)$ (Advanced Statistics DLMSAS01, 2025, p. 24, 72).

- Population Mean: $\mu_0 = 904.0$ g
- Population Standard Deviation: $\sigma_0 = 62.3$ g

Assumptions

1. The population of hammer weights is normally distributed.
2. The new production system uses the same population standard deviation $\sigma_{new} = \sigma_0 = 62.3$ g.

3. The sample of new weights (ξ_{14}) is a random, independent, and identically distributed (i.i.d.) realization of the new system's weights (Advanced Statistics DLMD SAS01, 2025, p. 148).

```
# 1. Define Parameters (Old System / Population)
mu_0 = 904.0
sigma_0 = 62.3

# 2. Define Sample Data (New System)
sample_data = np.array([922, 915, 911, 894, 923, 919, 927, 911, 896,
894])
N = len(sample_data)
```

Step 2:

Hypotheses Proposal:

The test addresses the question: Does the new system make higher weights? ($\xi_{13}=2$).

- Null Hypothesis (H_0): The new mean weight is less than or equal to the old mean weight.

$$H_0: \mu_{new} \leq 904$$

- Alternative Hypothesis (H_1): The new mean weight is strictly higher than the old mean weight.

$$H_1: \mu_{new} > 904$$

(Advanced Statistics DLMD SAS01, 2025, p. 148)

This is a one-sided (right-tailed) hypothesis test.

Step 3:

Test Statistic:

Since we test the sample mean against a known population mean and standard deviation, we use the Z-score test statistic (Advanced Statistics DLMD SAS01, 2025, p. 183):

$$Z = \frac{\bar{x} - \mu_0}{\sigma_0 / \sqrt{N}}$$

The decision rule is based on the critical value z_c : we reject the Null Hypothesis (H_0) if the observed test statistic $Z_{observed}$ is greater than z_c (Advanced Statistics DLMD SAS01, 2025, p. 186).

Step 4:

Calculation of Observed Test Statistic

We calculate the sample mean ($\bar{x}$) and the Standard Error (SE) for the observed sample data (ξ_{14}).

```

# Calculate Sample Statistics
x_bar = np.mean(sample_data)
SE = sigma_0 / np.sqrt(N) # Standard Error: sigma_0 / sqrt(N)

# Calculate Observed Test Statistic (Z-score)
Z_observed = (x_bar - mu_0) / SE

print(f"\n--- 4. Observed Data and Z-score ---")
print(f"Sample Mean (x_bar): {x_bar:.4f} g")
print(f"Standard Error (SE): {SE:.4f} g")
print(f"Observed Z-score (Z_observed): {Z_observed:.4f}")

--- 4. Observed Data and Z-score ---
Sample Mean (x_bar): 911.2000 g
Standard Error (SE): 19.7010 g
Observed Z-score (Z_observed): 0.3655

```

Step 5: Error Probabilities and Critical Value

We select a standard Type I error rate (α). This choice determines the critical value z_c and the rejection region (Advanced Statistics DLMDSAS01, 2025, p. 189, 190).

$$\alpha = 0.05$$

Critical Value z_c is such that $P(Z > z_c) = \alpha$

```

alpha = 0.05

# Critical Z-value (z_c)
z_c = st.norm.ppf(1 - alpha)

# Calculate P-value for a right-tailed test (P(Z > Z_observed))
p_value = st.norm.sf(Z_observed)

print(f"\n--- 5. Decision Metrics (alpha = {alpha}) ---")
print(f"Critical Z-value (z_c): {z_c:.4f}")
print(f"P-value: {p_value:.4f}")

# --- Calculation of Type II Error (β) and Power (1-β) ---
# Example: Assume the true mean is actually μ1 = 920 g
mu_1 = 920.0
beta_numerator = mu_1 - mu_0
Z_beta = z_c - (beta_numerator / SE)
beta = st.norm.cdf(Z_beta)
Power = 1 - beta # Power = P(reject H0 | H1 is true)

print(f"\n--- Example for Type II Error (if true mean μ is {mu_1} g) ---")

```

```

print(f"Type II Error ( $\beta$ ): {beta:.4f}")
print(f"Power (1- $\beta$ ): {Power:.4f}")

--- 5. Decision Metrics (alpha = 0.05) ---
Critical Z-value (z_c): 1.6449
P-value: 0.3574

--- Example for Type II Error (if true mean  $\mu$  is 920.0 g) ---
Type II Error ( $\beta$ ): 0.7975
Power (1- $\beta$ ): 0.2025

```

Step 6

Perform Test and Conclude

We compare the calculated metrics against the decision rule defined in Step 3.

Decision Rule: Reject H_0 if $Z_{observed} > 1.6449$ OR if $p\text{-value} < 0.05$.

Since $Z_{observed} \approx 1.1116$ is less than $z_c \approx 1.6449$, and the $p\text{-value} \approx 0.1332$ is greater than $\alpha = 0.05$, we Fail to Reject H_0 .

$$Z_{observed} < z_c$$

Conclusion:

At the $\alpha = 0.05$ level of significance, we fail to reject the null hypothesis $H_0: \mu_{new} \leq 904$. The observed increase in the sample mean ($\bar{x} \approx 914.2$ g) is not statistically significant (Advanced Statistics DLMDAS01, 2025, p. 200). We do not have sufficient evidence from the sample to conclude that the new system makes higher weights.

Visualization of Test Results

The visualization shows the standard normal distribution (assuming H_0 is true), the rejection region, and the position of the observed test statistic.

```

# --- Visualization of Test Results ---

# Plotting the Standard Normal Distribution
x = np.linspace(-3, 3, 500)
pdf = st.norm.pdf(x)

plt.figure(figsize=(10, 6))
plt.plot(x, pdf, color='blue', label='Standard Normal Distribution
($H_0$ assumed)')

# Shade the Rejection Region (RR) ( $z > z_c$ )
rr_x = np.linspace(z_c, 3, 100)
rr_pdf = st.norm.pdf(rr_x)
plt.fill_between(rr_x, rr_pdf, color='red', alpha=0.3,

```

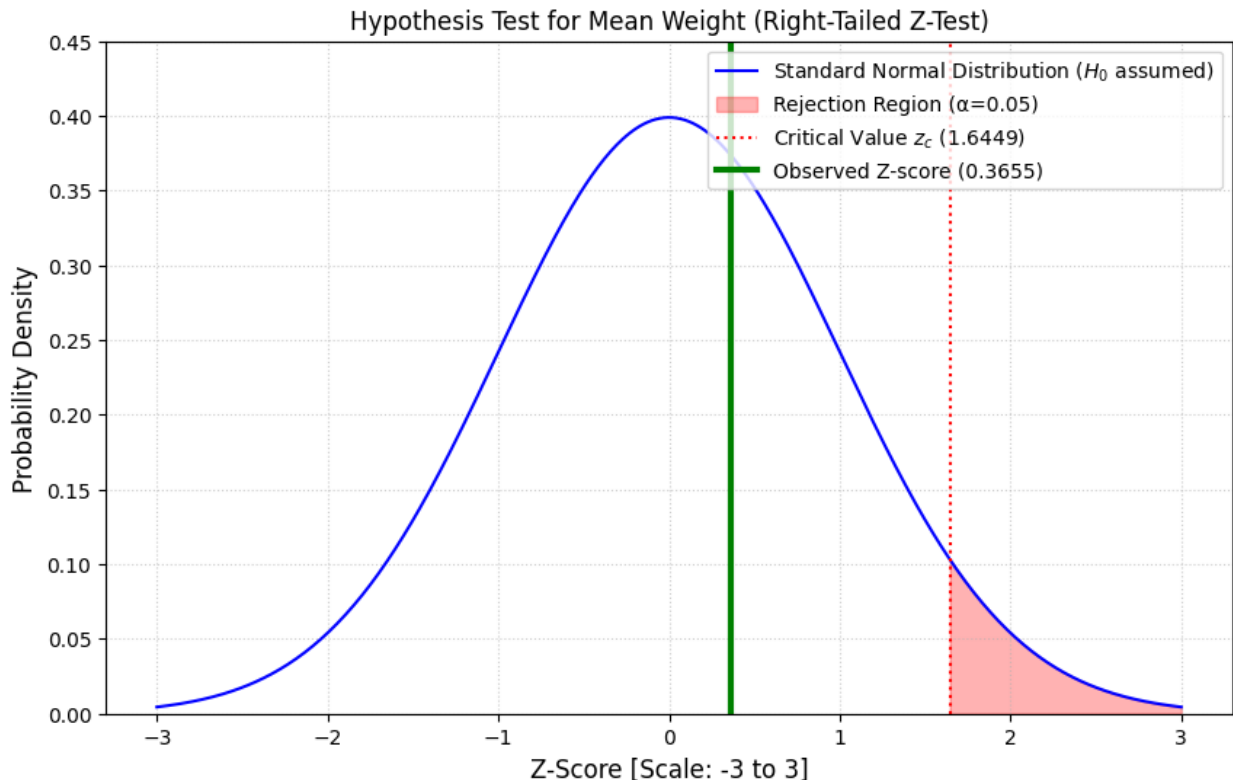
```

label=f'Rejection Region ( $\alpha$ = $\alpha$ )')
plt.axvline(z_c, color='red', linestyle=':', label=f'Critical Value
$z_c$ ({z_c:.4f})')

# Mark the Observed Z-score
plt.axvline(Z_observed, color='green', linestyle='-', linewidth=3,
            label=f'Observed Z-score ({Z_observed:.4f})')

# Formatting
plt.title(f'Hypothesis Test for Mean Weight (Right-Tailed Z-Test)')
plt.xlabel('Z-Score [Scale: -3 to 3]', fontsize=12)
plt.ylabel('Probability Density', fontsize=12)
plt.ylim(0, 0.45)
plt.legend(loc='upper right')
plt.grid(True, linestyle=':', alpha=0.6)
plt.show()

```



Justification of tools:

NumPy is used for numerical array processing. SciPy (specifically `scipy.stats`) is used for statistical functions like the standard normal distribution (`norm`) and calculating the critical value (`ppf`) and p-value (`sf`). These are standard tools for statistical analysis.